

SNAP-AI

AI-Powered Document Intelligence Platform

1. Project Overview

SNAP-AI is an AI-powered document intelligence platform that enables organizations to upload their documents, interact with them through natural-language conversation, and automatically extract meaningful insights onto dashboards. It is built as a multi-tenant system with support for multiple organizations, departments, and role-based access control.

2. Objective

- Eliminate the need to manually search through large volumes of organizational documents.
- Provide instant, context-grounded answers using Retrieval-Augmented Generation (RAG).
- Transform unstructured documents into structured, department-level insights and dashboards.

3. System Architecture

The platform follows a modular, service-oriented architecture composed of independent, loosely coupled components:

- **User Interface:** A responsive web application for chat, dashboards, document uploads, and administration.
- **Application Backend:** Handles authentication, organizations, departments, and core application data.
- **AI Processing Service:** A dedicated microservice for document processing, semantic search, and AI-generated responses.
- **Data & Vector Store:** A centralized database with integrated vector search that acts as the single source of truth.

4. How It Works

1. Document Upload

Users upload documents through the web interface.

2. Parsing & Chunking

Documents are parsed and split into smaller, manageable sections of text.

3. Semantic Indexing

Each section is converted into a numerical representation and stored for fast similarity search.

4. Query Understanding

A user's question is matched against the stored content to retrieve the most relevant sections.

5. Answer Generation

The retrieved context is passed to a large language model to produce accurate, grounded answers.

6. Persistence

Conversations and their context are saved so users can continue where they left off.

5. Key Features

- **Conversational Q&A:** Ask questions and receive answers grounded strictly in your own documents.
- **Automated Dashboards:** Automatically extracts and displays department-level metrics from documents.
- **Custom Metrics:** Users can define new dashboard metrics using simple natural-language prompts.
- **Multi-Tenant Access:** Organization- and department-level roles and permissions.
- **Duplicate Detection:** Prevents redundant uploads through content-based deduplication per organization.
- **Secure Onboarding:** Email verification during user signup.
- **Report Export:** Generates shareable output in multiple document formats.

6. Technology Highlights

- Modern web frontend for a fast, interactive user experience.
- Python-based backend and AI microservice architecture.
- Retrieval-Augmented Generation pipeline combining semantic search with a large language model.
- Cloud database with native vector search for scalable, intelligent retrieval.

7. Conclusion

SNAP-AI demonstrates a practical, production-oriented application of modern AI and Retrieval-Augmented Generation techniques. By combining semantic search, large language

models, and a secure multi-tenant architecture, it transforms static organizational documents into an interactive, insight-driven knowledge system.